

Design of a Distributed LoRa-TDMA Power Telemetry Network with On-Node AC Load Classification and a Precision Analog Front-End for Residential Energy Monitoring

Abstract—Residential electricity consumption in the Philippines remains poorly characterised at the appliance level. Commercially available smart meters provide aggregate kWh totals but offer no per-outlet load identification or real-time waveform data, limiting both consumer awareness and demand-side management programmes. This paper proposes a multi-node wireless power telemetry system that addresses these gaps through three integrated contributions. First, a precision analog front-end based on the ADS131M02 24-bit simultaneous-sampling delta-sigma ADC, a ZMPT101B galvanically isolated voltage transformer, and a salvaged toroidal current transformer (CT) replaces the commodity PZEM-004T v3 module. The new front-end provides simultaneous voltage and current sampling at 4 kSPS with programmable gain ($G=1$ to 128), full galvanic isolation from the AC mains, and sufficient bandwidth for harmonic analysis up to the 15th order. Second, an STM32F411 Cortex-M4F sensor node performs on-device harmonic feature extraction via a 128-point FFT and classifies connected loads into six residential categories—resistive, capacitive, single-phase motor, fan, switched-mode power supply, and lighting—using a deterministic decision tree with an upgradeable TFLite Micro multilayer perceptron (MLP). Third, an ESP32-S3 gateway coordinates up to eight sensor nodes through a frequency-hopping TDMA LoRa protocol with AES-128 CTR packet encryption, hosts a web-based single-page application (SPA) dashboard over Wi-Fi, and persists 120-point telemetry histories in ferroelectric RAM (FRAM). A GFSK bulk transfer extension enables over-the-air deployment of updated classifier models and calibration coefficient blocks within the existing superframe structure. The proposed system targets $\leq 0.5\%$ metering accuracy, $\geq 85\%$ classification accuracy across all six load categories, and $\geq 99\%$ uplink packet delivery ratio under residential indoor conditions.

Index Terms—LoRa, TDMA, load classification, non-intrusive load monitoring, ADS131M02, power quality, harmonic analysis, IoT, embedded machine learning, smart metering

I. INTRODUCTION

The residential sector accounts for a substantial and growing fraction of national electricity consumption in the Philippines, yet the granularity of available energy data remains coarse. Standard electromechanical and electronic energy meters report only aggregate kilowatt-hour totals, providing neither the per-outlet visibility needed for occupant feedback nor the appliance-level signatures required for automated demand management. Non-intrusive load monitoring (NILM) techniques [1] can, in principle, disaggregate whole-house

consumption into per-appliance contributions from a single metering point, but whole-house NILM requires significant signal processing resources and yields lower accuracy when many loads operate concurrently [2]. An alternative approach—monitoring individual branch circuits or outlets with networked smart plugs—trades installation complexity for classification accuracy, since each measurement point sees at most a few loads simultaneously.

Existing low-cost smart plug implementations typically rely on purpose-built power metering ICs such as the PZEM-004T v3, which integrates voltage sensing, current sensing, and RMS computation on a single module. While sufficient for energy auditing, these modules offer no access to the raw voltage or current waveform, precluding harmonic analysis and thereby ruling out any feature-based load classification. Replacing such a module with a discrete analog front-end and a high-resolution simultaneous-sampling ADC restores full waveform access at the cost of additional circuit complexity.

Long-range low-power wireless connectivity is a second challenge for multi-point metering. Wi-Fi and Bluetooth are well-supported in the Arduino and ESP-IDF ecosystems but impose high standby power and, for Wi-Fi, require AP association with its associated latency and join overhead. LoRa (Long Range), a chirp-spread-spectrum physical layer operating in unlicensed ISM bands, achieves kilometre-scale range at milliwatt transmit power and is increasingly used in sub-GHz IoT deployments [11]. A structured medium-access control layer is required, however, to coordinate multiple nodes without collision; frequency-division and time-division multiple access (TDMA) schemes have been demonstrated to be effective for deterministic, low-latency sensor networks [12].

Security is a practical concern for metering infrastructure. An unencrypted LoRa broadcast reveals occupancy patterns, appliance usage schedules, and potentially security-relevant information (vacation periods, nighttime routines). AES-128 in counter (CTR) mode provides a lightweight, nonce-based stream cipher that is compatible with embedded C++ without dynamic memory allocation and is recommended by NIST [15].

This paper proposes a system that integrates all three of these requirements: precision analog measurement, structured wireless MAC, and load classification—within a hard-

ware/firmware architecture that is practical to build and deploy in a Philippine residential setting.

A. Objectives

The specific objectives of this research are:

- 1) Design and prototype a precision analog front-end using the ADS131M02 24-bit simultaneous-sampling ADC with a ZMPT101B isolated voltage transformer and a toroidal CT, targeting $\leq 0.5\%$ metering accuracy and full galvanic isolation from the AC mains.
- 2) Implement a TDMA LoRa communication protocol with frequency-hopping channel access, AES-128 CTR packet encryption, and a GFSK bulk transfer extension for over-the-air model and calibration deployment.
- 3) Develop an on-node harmonic feature extraction pipeline using a 128-point FFT and derive a 10-dimensional feature vector for load identification without transmitting raw waveform data.
- 4) Implement and evaluate a two-stage on-node classifier: a deterministic decision tree for immediate deployment and a compact TFLite Micro MLP trained on collected field data, targeting $\geq 85\%$ accuracy across six residential load categories.
- 5) Develop an ESP32-S3 gateway with a web-based SPA dashboard for real-time visualization, remote relay control, persistent energy history via FRAM, and over-the-air classifier management.
- 6) Validate the integrated system against reference measurements, characterizing metering accuracy, classification accuracy, and uplink packet delivery ratio under realistic residential conditions.

B. Scope and Limitations

The system targets single-phase 220 V / 60 Hz residential loads of up to 30 A per monitored outlet. Gateway-to-internet connectivity (cloud backhaul) is out of scope; the dashboard is self-hosted on the gateway node. Load disaggregation of mixed-load outlets is not addressed. The analogue front-end is designed for indoor perfboard prototype construction; PCB layout and EMC compliance are deferred.

II. RELATED WORK

A. Non-Intrusive Load Monitoring and Load Classification

Hart's foundational work [1] established the framework for identifying appliances from changes in steady-state current and power at a single metering point. The approach works well for stepwise loads but struggles with variable-speed drives and modern switched-mode supplies. Subsequent work extended feature sets to include harmonic content [3], transient event detection, and V-I trajectory analysis [4]. Zeifman and Roth [2] survey the state of the art and identify harmonic-ratio features as among the most discriminative for residential load types.

Data-driven approaches have gained traction as labelled datasets became available. Kolter and Johnson [5] released the REDD dataset and demonstrated that hidden Markov

models can exceed the accuracy of threshold-based disaggregation. Kelly and Knottenbelt [6] applied recurrent neural networks directly to raw power traces, achieving state-of-the-art disaggregation on the UK-DALE dataset. For per-outlet classification (as opposed to whole-house disaggregation), the classification problem is considerably simpler because each metering point typically serves one or a few loads; hand-crafted feature-based classifiers have been shown to reach $> 90\%$ accuracy in this setting [7].

The proposed system adopts a hybrid approach: a deterministic decision tree provides immediate classification without training data, while a compact MLP—deployed via OTA after supervised data collection—improves accuracy on atypical or region-specific loads.

B. Precision Power Measurement with Sigma-Delta ADCs

High-resolution simultaneous-sampling ADCs have become the preferred architecture for precision power quality instruments. The ADS131M02 (Texas Instruments) [9] is a 24-bit, dual-channel, simultaneously sampling delta-sigma ADC with an on-chip PGA (gain 1–128) and a sinc3 decimation filter, delivering 18–20 effective bits at output data rates of 1–32 kSPS. The simultaneous sampling architecture eliminates the phase skew inherent in multiplexed ADCs, which is critical for accurate reactive-power and power-factor measurement.

Several precision metering IC designs reported in the literature use this class of ADC. Adamo et al. [8] describe a power quality monitor based on a 16-bit simultaneous ADC, demonstrating that waveform-based harmonic analysis is practical in embedded systems. The ZMPT101B miniature current-mode voltage transformer provides 4 kV galvanic isolation from the AC mains with less than 20 arcminutes of phase error at the rated operating current, making it a cost-effective alternative to resistive dividers for safe isolated voltage sensing [10].

C. LoRa and LPWAN for IoT Sensing

LoRa modulation, developed by Semtech [14], uses chirp spread spectrum (CSS) over configurable spreading factors (SF6 to SF12), trading data rate for link margin. At SF6 the air-time for a 32-byte packet is approximately 10–15 ms with ~ 10 dB less processing gain than SF12; this is appropriate for TDMA networks where short slot durations are required. Augustin et al. [11] characterise LoRa coverage in urban and indoor environments, confirming reliable indoor coverage at 20–30 m through multiple concrete walls at SF6 with 10 dBm transmit power.

TDMA-based MAC layers for LoRa have been proposed to overcome the collision-prone LoRaWAN ALOHA-like access scheme. Bor et al. [13] show that LoRaWAN does not scale beyond a few hundred devices in dense deployments, motivating structured MAC protocols for high-density or latency-sensitive applications. Frequency hopping across multiple channels increases resilience against narrowband interference and partial-band fading [12].

[System Architecture Diagram]
 Gateway (ESP32-S3) ← LoRa TDMA → Nodes × 8
 (STM32F411 + ADS131M02)
 Gateway ← Wi-Fi → Web Dashboard

Fig. 1. Proposed system architecture. The ESP32-S3 gateway acts as the TDMA master and Wi-Fi access point. Up to eight STM32F411 sensor nodes report simultaneously sampled voltage and current waveforms and classified load types every 3 s superframe.

D. Embedded Machine Learning for IoT

The TFLite Micro runtime [18] enables inference from quantised TensorFlow Lite models on microcontrollers with as little as 16 kB of RAM. Warden and Situnayake [19] survey the domain and demonstrate that small (hundreds of parameters) multilayer perceptrons achieve adequate accuracy for many IoT classification tasks while executing in under 1 ms on Cortex-M4 processors. The ARM CMSIS-DSP library [20] provides optimised FFT and vector operations for Cortex-M processors, enabling efficient feature extraction without hand-tuned assembly.

E. Security in LoRa IoT Networks

LoRaWAN mandates AES-128 for join-request authentication and payload encryption, demonstrating that AES is a practical security primitive for constrained IoT devices [16]. NIST SP 800-38A [15] defines CTR mode, which converts a block cipher into a stream cipher suitable for variable-length packets and allows parallelised encryption without IV re-use if the nonce is unique per packet. Hardware RFID-based key provisioning, as studied by Hancke and Kuhn [17], provides a physical-world complement to over-the-air key distribution.

III. SYSTEM ARCHITECTURE

Fig. 1 illustrates the proposed system. Two hardware roles share a single firmware codebase compiled under different PlatformIO build environments.

Gateway (ESP32-S3): The gateway is built on the Espressif ESP32-S3 SoC (Xtensa LX7 dual-core at 240 MHz, 512 kB SRAM). It acts as the TDMA master, transmitting a beacon at the start of each 3 s superframe and allocating uplink/downlink slots to registered nodes. A Wi-Fi STA/AP interface serves the management dashboard. An MB85RC256V 32 kB FRAM chip (I²C, 0x50) persists per-node energy accumulators and 120-point telemetry histories across reboots. An optional PN532 NFC reader supports hardware-based AES key provisioning in encrypted builds.

Sensor Nodes (STM32F411): Each sensor node uses the STMicroelectronics STM32F411 “Black Pill” (ARM Cortex-M4F at 100 MHz, 128 kB SRAM). The node samples the AC mains via the custom analog front-end (Section IV-A), runs the on-node load classifier (Section IV-D), controls a relay for switched load management, and transmits a 32-byte telemetry packet in its assigned TDMA uplink slot.

Radio Subsystem: Both gateway and nodes use the Semtech SX1278 LoRa transceiver connected via SPI. The gateway uses the standard ESP32-S3 FSPI peripheral; the nodes use STM32 SPI1 for the ADS131M02 ADC and SPI2 for the SX1278. The RadioLib library [21] abstracts low-level transceiver control.

IV. METHODOLOGY

A. Analog Front-End Design

1) *Current Channel:* The current channel uses the CT bundled with the PZEM-004T module—a solid-core toroidal transformer with a 1:1000 turns ratio and 100 A primary rating. The CT secondary drives a split burden resistor ($2 \times 5 \Omega$, 0.1%, 15 ppm/°C metal film), whose center tap is connected to AGND. This balanced configuration eliminates even-order distortion in the ADC input stage.

At the 30 A target load, the peak burden voltage is:

$$V_{b,pk} = \frac{I_{max}\sqrt{2}}{n} \cdot R_b = \frac{30\sqrt{2}}{1000} \times 10 = 0.424 \text{ V}, \quad (1)$$

corresponding to 70.7% of the ADS131M02 full-scale range at PGA gain $G = 2$ (± 0.6 V). The clipping threshold at $G = 2$ is 42.4 A RMS, accommodating steady-state residential loads; motor inrush transients up to 84.9 A RMS are captured by switching to $G = 1$ in firmware. The noise-limited current resolution at 4 kSPS ($G = 2$, $e_n \approx 3.5 \mu\text{V}$) is:

$$\Delta I_p = \frac{e_n}{R_b} \cdot n = \frac{3.5 \times 10^{-6}}{10} \times 1000 = 0.35 \text{ mA}, \quad (2)$$

yielding a dynamic range of approximately 60 000:1, far exceeding the 5 000:1 design target.

2) *Voltage Channel:* The ZMPT101B miniature current-mode voltage transformer (1:1 ratio, 4 kV isolation) senses the mains voltage. Two 56 k Ω series resistors ($R_L = 112 \text{ k}\Omega$ total) limit the primary operating current to 2 mA at 220 V RMS. The secondary current flows through a split sampling resistor ($2 \times 160 \Omega$, center tap to AGND), producing a peak differential output of 0.917 V at 220 V RMS—76.4% of the ADS131M02 full-scale at $G = 1$. The ZMPT101B introduces a phase lag of $\leq 0.33^\circ$, characterised in [10].

3) *Anti-Aliasing Filter and Phase Calibration:* Both channels share an identical differential RC anti-aliasing filter: $R_{filt} = 100 \Omega$ on each line, $C_{diff} = 680 \text{ nF}$ COG ceramic across the differential pair, giving:

$$f_c = \frac{1}{2\pi R_{filt} C_{diff}} = \frac{1}{2\pi \times 100 \times 680 \times 10^{-9}} = 2341 \text{ Hz}. \quad (3)$$

Attenuation at the fundamental (60 Hz) is 0.003 dB; at the 15th harmonic (900 Hz) it is 0.59 dB—a systematic, known,

TABLE I
PGA GAIN-RANGING STATES – CURRENT CHANNEL

Gain	FSR ($\pm V$)	Clip (A RMS)	Noise Floor (mA)
1	1.20	84.9	0.53
2	0.60	42.4	0.35
4	0.30	21.2	0.20
16	0.075	5.3	0.095

and correctable gain error. Because both channels use identical filter components, the differential phase error contributed by the anti-aliasing filter is exactly zero [9]; the ADS131M02's simultaneous sampling eliminates ADC-induced phase skew. Residual differential phase error from the ZMPT101B and CT transformers is corrected via the ADS131M02 on-chip PHASE register at a resolution of:

$$\Delta\phi_{\text{step}} = 360^\circ \times f_{\text{mains}} \times \frac{1}{f_{\text{CLKIN}}} = 360 \times 60 \times \frac{1}{8.192 \times 10^6} \approx 0.00264^\circ \quad (4)$$

4) *PGA Gain-Ranging*: Four gain states are defined in firmware (Table I). At each 60 Hz cycle boundary, the firmware checks whether the peak sample amplitude exceeds 90% of the current FSR (switch to lower gain) or falls below 15% (switch to higher gain). Gain transitions are applied via SPI register write and take effect on the next ADS131M02 conversion. The strategy maintains high ADC utilisation across a 5000:1 current range without increasing burden resistance.

5) *Firmware Integration on STM32F411*: The ADS131M02 DRDY pin is connected to an EXTI interrupt (priority 0, highest). The ISR launches a DMA-driven SPI1 read of the 80-bit frame (status word, 24-bit Ch0, 24-bit Ch1, CRC-16). A DMA completion callback copies the two signed 24-bit samples into circular buffers, waking the `adcTask` (FreeRTOS, STM32duino) for per-sample accumulation. At 4 kSPS the total CPU load from ADC acquisition and per-sample arithmetic is approximately 2.2%, leaving > 97% idle for the TDMA radio task and classifier.

B. TDMA Protocol and Wireless Communication

1) *Superframe Structure*: The network organises time into 3000 ms superframes (Fig. 2). Each superframe contains:

- A 40 ms beacon (gateway $\rightarrow$ all nodes) on Ch 0 at SF6, 125 kHz BW.
- Eight 165 ms data slots (DL 50 ms + UL 100 ms + 15 ms guard) for eight registered nodes.
- A 60 ms contention window (uplink) and a 40 ms contention window (downlink) for unregistered nodes to join.
- A 1520 ms idle window used for gateway maintenance and, when a bulk transfer is active, a GFSK burst phase.
- A 20 ms end guard before the next beacon.

Downlink-first slot ordering gives the gateway the opportunity to deliver commands (relay toggle, schedule, threshold) immediately before the node's uplink. The node's uplink slot timing is derived from the beacon receive timestamp:

$$t_{\text{tx}} = t_{\text{bcn}} + (T_{\text{bcn}} - T_{\text{air}}) + (k - 1) \cdot T_{\text{slot}} + T_{\text{DL}}, \quad (5)$$

[Superframe timing diagram:
Beacon (40ms) — Slots 1-8 (8x165ms=1320ms) —
CW-UL (60ms) — CW-DL (40ms) — Idle (1520ms) —
Guard (20ms)]

Fig. 2. TDMA superframe structure. The 1520 ms idle window is normally quiescent but is repurposed for GFSK bulk data transfers when a bulk session is active.

where $T_{\text{bcn}} = 40$ ms, $T_{\text{air}} = 18$ ms (on-air time of the 8-byte beacon at SF6), $T_{\text{slot}} = 165$ ms, $T_{\text{DL}} = 50$ ms, and k is the node's assigned slot index (1–8).

2) *Frequency Hopping*: The SX1278 is configured with eight 200 kHz-spaced channels in the 433 MHz ISM band (Ch 0 = 433.050 MHz, Ch 7 = 434.450 MHz). The active channel for each slot within a superframe is:

$$\text{ch} = (\text{sfCount} \times 7 + k) \bmod 8, \quad (6)$$

where *sfCount* is the superframe counter and the coprime multiplier 7 ensures that every slot visits all eight channels across eight consecutive superframes. Beacon and contention windows use Ch 0. The hopping sequence is deterministic and requires no explicit signalling.

3) *Node Lifecycle and Network Epoch*: Unregistered nodes transmit a `JoinRequest` (0xA0, 4 bytes) during the contention window. The gateway allocates the first available slot and replies with a `JoinAck` (0xA1, 4 bytes) containing the assigned slot index and the current network epoch. A `uint8_t` epoch field in the beacon increments on every node eviction (which occurs only when all 8 slots are occupied and a node has missed 8 consecutive superframes ≈ 24 s). A registered node that detects an epoch mismatch immediately re-contents rather than transmitting in a potentially reassigned slot.

4) *Packet Encryption*: When the `PKT_ENCRYPTION` build flag is active, all uplink and downlink packets (excluding the plaintext *sfCount* bytes of the beacon) are encrypted in-place using AES-128 in CTR mode [15]. The 16-byte nonce is constructed as:

$$\text{nonce} = [\text{sfCount}_{\text{lo}} \mid \text{sfCount}_{\text{hi}} \mid \text{slotId} \mid \text{dir} \mid \mathbf{0}_{12}], \quad (7)$$

where *dir* is a per-packet-type direction discriminator byte. This guarantees a unique keystream block per (superframe, slot, direction) triple. The AES key is auto-generated by the gateway on first boot and provisioned to nodes via a PN532 I²C NFC reader using MIFARE Classic sector 1 (Key A 0xFF×6) as described in [17].

C. GFSK Bulk Transfer Extension

Several subsystems require payloads of 28–4096 bytes that cannot be delivered efficiently through the 7-byte downlink

[Bulk transfer timing diagram:
 ...CW-DL (40ms) — Maint (20ms) — LoRa- ζ GFSK —
 GFSK Burst — GFSK- ζ LoRa — Idle — Guard]

Fig. 3. GFSK bulk transfer phase within the idle window. TDMA data slots and contention windows are completely unaffected.

slot window: ADC calibration coefficient blocks (24 bytes), classifier threshold sets (28 bytes), quantised TFLite Micro MLP models (≈ 500 bytes), and training-data upload batches (2–4 kB). Delivering a 500-byte payload through the 7-byte DL window would require 72 superframes (216 s), which is impractical.

The GFSK bulk transfer extension repurposes the 1520 ms idle window for a high-throughput GFSK burst, while leaving all eight TDMA data slots entirely undisturbed (Fig. 3). The protocol is application-agnostic: it delivers an opaque byte buffer between the gateway and one target node per superframe.

1) *Physical Layer*: The SX1278 supports FSK/GFSK packet mode alongside LoRa through the LongRangeMode bit in RegOpMode. Mode switching requires a transition through Sleep mode and takes approximately 0.5 ms. GFSK is configured at 100 kbps bit rate, 50 kHz frequency deviation (modulation index $\beta = 1.0$), 125 kHz single-sided RX bandwidth, BT = 0.5 Gaussian filtering, and CRC-16/CCITT hardware error detection. At 100 kbps, a worst-case 4096-byte transfer (21 fragments of 200 bytes each) completes in approximately 462 ms, well within the 1476 ms usable bulk window.

The indoor link budget at 433 MHz over 20 m through two concrete walls is estimated at 73 dB path loss. With 10 dBm transmit power and -104 dBm sensitivity at 100 kbps, the margin is 41 dB—adequate for body shadowing and transient fading [11].

2) *Fragment Protocol*: The bulk payload is segmented into 200-byte fragments, each encapsulated in a BulkFragHeader (pktType 0xB0, sequence number, fragment count, payload length) followed by the payload and a hardware CRC-16. A stop-and-wait ARQ scheme with a 50 ms ACK timeout and up to 3 retransmissions per fragment provides reliable delivery. An end-to-end CRC-32 over the complete reassembled payload detects any reassembly error that per-fragment CRC would miss.

In encrypted builds, each fragment uses an extended nonce incorporating the fragment sequence number in byte 4, guaranteeing unique keystreams across fragments within a transfer

TABLE II
FEATURE VECTOR DEFINITION

Index	Symbol	Definition
0	f_{PF}	Displacement power factor
1	f_{Qsgn}	Sign of reactive power ($\pm 1, 0$)
2	f_{THD}	$THD_I = \sqrt{\sum I_h ^2} / I_1 $
3	f_{H3}	$ I_3 / I_1 $
4	f_{H5}	$ I_5 / I_1 $
5	f_{H7}	$ I_7 / I_1 $
6	f_{CF}	Crest factor I_{pk} / I_{rms}
7	f_{OE}	$(I_2 + I_4) / I_1 $ (asymmetry)
8	f_{H53}	$ I_5 / I_3 $ (shape discriminator)
9	f_{Prms}	$\log_2(I_{rms}/0.05)$, clamped [0, 10]

and across retried transfers in different superframes:

$$\text{nonce}_{\text{bulk}} = [sfCount_{lo} \mid sfCount_{hi} \mid slotId \mid dir \mid seqNum \mid \mathbf{0}_{11}]. \quad (8)$$

The gateway signals the start of a bulk session by inserting a 6-byte BulkGrantPacket (pktType 0x08) into the target node's regular LoRa downlink slot. The node acknowledges readiness by setting bit 7 of the statusFlags byte in its subsequent uplink telemetry packet. If the gateway observes this flag, both sides simultaneously switch to GFSK at the computed bulk phase start time; otherwise, the grant is retried in the next superframe.

D. Feature Extraction and Load Classification

1) *Feature Extraction Pipeline*: Every 2 s (approximately 120 mains cycles at 60 Hz), the adcTask on the STM32F411 applies a 128-point FFT to the most recent current waveform samples. With $f_s = 4000$ SPS and $N = 128$, the FFT frequency resolution is:

$$\Delta f = \frac{f_s}{N} = \frac{4000}{128} = 31.25 \text{ Hz}. \quad (9)$$

The fundamental (60 Hz) maps to bin $k_1 = 2$; odd harmonics up to the 15th (900 Hz) map to bins 2, 6, 10, 13, 17, 21, 25, 29. A Hann window reduces spectral leakage; since only magnitude ratios are used for classification, the 6 dB main-lobe broadening is acceptable. The anti-aliasing filter's known attenuation at each harmonic frequency is compensated by the inverse filter magnitude:

$$|I_{h,\text{corr}}| = |I_h| \times \sqrt{1 + \left(\frac{h \cdot 60}{f_c}\right)^2}. \quad (10)$$

From the corrected harmonic magnitudes, a 10-dimensional feature vector $\mathbf{x} = [f_0, \dots, f_9]$ is computed (Table II). All features are dimensionless ratios or normalised quantities, making the classifier independent of absolute load magnitude. The total feature vector occupies 40 bytes.

2) *Six-Class Load Taxonomy*: Six load categories are defined to cover the dominant residential load types in a 220 V / 60 Hz Philippine residential installation (Table III). The taxonomy is designed so that each class has at least one

TABLE III
SIX-CLASS LOAD TAXONOMY

ID	Class	Typical Loads
0	RESISTIVE	Incandescent, flat iron, water heater
1	CAPACITIVE	PFC-corrected SMPS, UPS on battery
2	MOTOR_1PH	Aircon compressor, refrigerator
3	FAN	Ceiling fan, desk fan, exhaust fan
4	SMPS	Laptop/phone charger, desktop PSU
5	LIGHTING	CFL, LED lamp (non-PFC), fluorescent

[Decision Tree Diagram:
Root: $I_{rms}; 50\text{mA}$? $\rightarrow$ RESISTIVE
THD $_I > 0.15$? $\rightarrow$ Low-THD branch (PF, Q_{sgn}) / High-THD branch (CF, H3)]

Fig. 4. Simplified decision tree structure. Each threshold is stored in a ClassifierThresholds struct and can be updated via OTA bulk transfer from the gateway.

strongly discriminative feature: SMPS loads are identified by their high crest factor (> 2.2); lighting (CFL/LED) by high H3/H1 ratio (> 0.35); motor loads by lagging power factor with low THD; fans by deep lagging power factor; capacitive loads by negative reactive power; and resistive loads by near-unity power factor and low THD.

3) *Baseline: Deterministic Decision Tree*: The primary classifier is a hand-crafted decision tree that requires no labelled training data, executes in under $1 \mu\text{s}$ on the STM32F411, and is fully interpretable. The tree follows a “peel off the easy cases first” strategy (Fig. 4):

- 1) **No-load gate** ($I_{rms} < 50\text{mA}$): below the noise floor; classify as RESISTIVE (idle).
- 2) **THD split** ($\text{THD}_I < 0.15$): separates approximately sinusoidal loads (motor, fan, resistive, capacitive) from distorted loads (SMPS, lighting).
- 3) **Low-THD branch**: $\text{PF} > 0.95 \Rightarrow \text{RESISTIVE}$; $Q_{sgn} < 0 \Rightarrow \text{CAPACITIVE}$; $\text{PF} > 0.72 \Rightarrow \text{MOTOR_1PH}$; else FAN.
- 4) **High-THD branch**: $\text{CF} > 2.2 \Rightarrow \text{SMPS}$; $\text{H3/H1} > 0.35 \Rightarrow \text{LIGHTING}$; residual: apply PF and Q_{sgn} as in step 3.

A confidence score (0–7, packed into 3 bits) is computed

TABLE IV
GATEWAY REST AND WEBSOCKET API (SELECTED)

Endpoint	Function
GET /api/nodes	All registered node states
GET /api/node/{id}/history	120-point V/I/P/class history
POST /api/bulk	Enqueue GFSK bulk DL/UL transfer
GET /api/features	Active build-flag feature set
POST /api/login	Issue session token
WS: telemetry	Real-time node update push
WS: relay_manual	Toggle relay on/off
WS: relay_schedule	Set timed relay schedule
WS: set_threshold	Update power-off threshold

as the normalised distance from the crossed threshold. A 5-sample temporal smoother (confidence-weighted majority vote over 10 s of history) suppresses transient misclassifications during motor startup. The complete classification result, including confidence and transient flag, is packed into a single byte within the 32-byte TelemetryPacket.

4) *Upgrade Path: TFLite Micro MLP*: After supervised training data are collected (Phase 1, Section V), the decision tree is replaced by a compact 3-layer MLP (10 inputs, 16 hidden (ReLU), 12 hidden (ReLU), 6 outputs (softmax)) with 458 parameters. Post-training 8-bit quantisation reduces the model to approximately 500 bytes, deployable over the GFSK bulk channel. The TFLite Micro runtime requires a 4 kB tensor arena; inference takes approximately $2 \mu\text{s}$ on the STM32F411 Cortex-M4F using CMSIS-DSP-accelerated matrix operations [19], [20]. A fallback mechanism reverts to the decision tree when the MLP softmax probability is below 0.5.

E. Gateway Dashboard and Data Management

The ESP32-S3 gateway hosts a LittleFS single-page application (SPA) comprising HTML, CSS, JavaScript, and Chart.js, served via the ESPAsyncWebServer library. The SPA communicates with the gateway through a REST API and a WebSocket endpoint. Table IV summarises the key REST endpoints.

Per-node energy accumulators, 120-point telemetry histories (encoded as HistoryPoint structs at 16 bytes each), and node labels are persisted in the MB85RC256V FRAM. The memory map allocates 1968 bytes per node slot (8 slots), occupying 15.7 kB of the 32 kB chip. FRAM writes are deferred through a dirty-count mechanism (threshold 10) and dispatched to a dedicated Core 0 FRAM task via a FreeRTOS queue, ensuring that I²C writes never block the Core 1 TDMA timing task. Cross-slot UID lookup allows the gateway to restore a node’s energy history and label after a reboot even if the node is assigned a different slot index.

The dashboard password is stored in ESP-IDF NVS (namespace wifi-cfg, key dashpass). When a password is set, all REST and WebSocket endpoints require a session token obtained from POST /api/login. The system also supports optional AP password protection, NTP-synchronized timestamps, and static IP configuration.

TABLE V
LOAD CLASSIFICATION VALIDATION TEST SET

Class	Physical Test Loads
RESISTIVE	100 W incandescent, electric flat iron, space heater
CAPACITIVE	Active-PFC laptop charger, UPS (battery mode)
MOTOR_1PH	0.75 HP window aircon, refrigerator compressor, washing machine
FAN	75 W ceiling fan, desk fan, exhaust fan
SMPS	Laptop charger (no PFC), desktop PC PSU, 5 W USB charger
LIGHTING	13 W CFL, 9 W LED bulb (non-PFC), 36 W fluorescent + magnetic ballast

F. System Integration

The firmware is compiled from a single codebase under ten PlatformIO environments differentiating gateway vs. node role, encrypted vs. plaintext, hardware PZEM vs. simulated sensor, and ESP32-dev vs. ESP32-S3 vs. STM32F411 target. Shared packet structures (TelemetryPacket, BeaconPacket, etc.) are defined once in `tdma_protocol.h` with `#pragma pack(1)`.

A hardware abstraction layer (HAL) in `src/hal/` wraps all platform-exclusive APIs (reboot, RNG, NVS, AES, task creation, spinlock) behind thin C function headers. Application code above the HAL is platform-agnostic; only the four `hal/*.cpp` implementation files differ between the ESP32 and STM32 targets.

V. VALIDATION PLAN

A. Phase 1: Analog Front-End Characterisation

The analog front-end will be characterised against a calibrated Fluke 435-II power quality analyser as the reference. Test conditions cover:

- Voltage accuracy: mains voltage varied $\pm 10\%$ around 220 V via a variac; target $\leq 0.5\%$ error.
- Current accuracy: precision resistive loads (100 W bulb, 1000 W heater) measured at PF = 1.0; target $\leq 0.5\%$ error.
- Power factor accuracy: ceiling fan and window-type aircon (PF = 0.55–0.90); target ≤ 0.005 .
- Phase calibration: empirical determination of CT phase error at 5 A and 30 A load levels using the two-point PHASE register correction procedure.

B. Phase 2: Load Classification Validation

The decision tree classifier will be validated against 18 physical loads (3 examples per class, listed in Table V). Acceptance criteria: $\geq 85\%$ class-averaged accuracy; no single class below 70%. After validation, supervised labelling data will be collected (50–100 feature vectors per load) and used to train the TFLite Micro MLP offline. The trained model will be deployed via the GFSK bulk channel and re-evaluated against the same test set.

C. Phase 3: Network Performance Characterisation

With all 8 node slots active and transmitting simulated telemetry (PZEM_FAKE build flag), the uplink packet delivery ratio (PDR) will be measured over 1000 consecutive superframes (50 min) in a residential indoor environment (concrete walls, kitchen/living room separation) at typical node-to-gateway distances of 5–20 m. Target: PDR $\geq 99\%$. Latency from sensor sampling to gateway WebSocket push will be measured; target: < 3.5 s (one superframe cycle).

D. Phase 4: GFSK Bulk Transfer Validation

The GFSK bulk transport will be validated through:

- 1) Echo test (DL + UL round-trip) at payload sizes 1 B, 200 B, 201 B, 500 B, and 4096 B.
- 2) Forced-retransmission test: deliberate NACK injection to verify ARQ recovery and end-to-end CRC-32 integrity.
- 3) Overrun protection: verify the GFSK phase terminates before `END_GUARD_MS` under maximum payload + maximum retries.
- 4) Telemetry PDR regression: PDR must remain $\geq 99\%$ during bulk-active superframes.

E. Phase 5: MLP Deployment and Re-Evaluation

The quantised TFLite Micro MLP model (≈ 500 bytes) will be deployed to all 8 nodes via the GFSK broadcast mechanism (`POST /api/bulk/broadcast`). Classification accuracy will be re-measured using the same 18-load test set. Target: $\geq 90\%$ class-averaged accuracy with the MLP, compared to the decision tree baseline.

VI. EXPECTED CONTRIBUTIONS AND OUTCOMES

The proposed system makes the following contributions:

- 1) **Precision waveform-accessible metering:** The ADS131M02 + ZMPT101B + CT front-end provides simultaneous voltage and current sampling at 4 kSPS with $\leq 0.5\%$ accuracy and full galvanic isolation, enabling harmonic analysis that is impossible with the PZEM-004T.
- 2) **Scalable deterministic MAC:** The frequency-hopping TDMA protocol supports up to 8 nodes in a 3 s superframe with guaranteed slot timing, collision-free operation, and a provably unique per-packet keystream for AES-128 CTR encryption.
- 3) **On-node, no-transmission-cost load classification:** The decision tree and MLP classifiers run entirely on the STM32F411 node; only a 1-byte result is transmitted per superframe. This avoids the latency and bandwidth cost of raw waveform uplink and maintains classification capability even when the gateway is unreachable.
- 4) **OTA model management:** The GFSK bulk transfer extension enables deployment of updated classifier models and calibration data to all nodes within 24 s without disrupting ongoing telemetry, providing a practical path from decision-tree to trained MLP operation.
- 5) **Self-hosted residential dashboard:** The web SPA requires no cloud infrastructure; the gateway operates as

an access point or connects to an existing router, making the system deployable in locations without internet connectivity.

The system is intended to serve as a research platform for evaluating harmonic-signature load classification under Philippine residential conditions (220 V / 60 Hz, tropical climate, diverse load mix). Data collected through the supervised labelling mode will contribute to a small labelled power signature dataset for the Philippine residential context, where public datasets such as REDD [5] and UK-DALE [6] are not directly applicable due to differing mains frequency (60 Hz vs. 50 Hz) and appliance profiles.

VII. CONCLUSION

This paper has proposed a distributed power telemetry and load classification system tailored for Philippine residential energy monitoring. The system replaces the commodity PZEM-004T module with a precision ADS131M02 analog front-end that provides simultaneous voltage/current sampling, full galvanic isolation, and PGA gain-ranging across a 5000:1 current dynamic range. A TDMA LoRa MAC with frequency hopping, AES-128 CTR encryption, and a novel GFSK bulk transfer extension provides scalable, secure wireless connectivity for up to 8 nodes. An on-node harmonic feature extraction pipeline feeding a deterministic decision tree (with a TFLite Micro MLP upgrade path) classifies connected loads into six residential categories without transmitting raw waveform data.

The integrated system is designed to validate $\leq 0.5\%$ metering accuracy, $\geq 85\%$ classification accuracy, and $\geq 99\%$ packet delivery ratio under realistic residential conditions. The proposed work addresses gaps in existing low-cost metering and NILM literature by combining precision simultaneous-channel ADC metering, structured encrypted wireless MAC, and embedded load classification in a single self-hosted platform applicable to the 60 Hz Philippine residential context.

REFERENCES

- [1] G. W. Hart, "Nonintrusive appliance load monitoring," *Proc. IEEE*, vol. 80, no. 12, pp. 1870–1891, Dec. 1992.
- [2] M. Zeifman and K. Roth, "Nonintrusive appliance load monitoring: Review and outlook," *IEEE Trans. Consum. Electron.*, vol. 57, no. 1, pp. 76–84, Feb. 2011.
- [3] W. Wichakool, A.-T. Avestruz, R. W. Cox, and S. B. Leeb, "Modeling and estimating current harmonics of variable electronic loads," *IEEE Trans. Power Electron.*, vol. 24, no. 12, pp. 2803–2811, Dec. 2009.
- [4] H. Y. Lam, G. S. K. Fung, and W. K. Lee, "A novel method to construct taxonomy of electrical appliances based on load signatures," *IEEE Trans. Consum. Electron.*, vol. 53, no. 2, pp. 653–660, May 2007.
- [5] Z. Kolter and M. Johnson, "REDD: A public data set for energy disaggregation research," in *Proc. Workshop Data Mining Appl. Sustain. (SustKDD)*, 2011.
- [6] J. Kelly and W. Knottenbelt, "Neural NILM: Deep neural networks applied to energy disaggregation," in *Proc. 2nd ACM Int. Conf. Embedded Syst. Energy-Efficient Built Environ. (BuildSys)*, 2015, pp. 55–64.
- [7] O. Parson, S. Ghosh, M. Weal, and A. Rogers, "An unsupervised training method for non-intrusive appliance load monitoring," *Artif. Intell.*, vol. 217, pp. 1–19, Dec. 2014.
- [8] F. Adamo, F. Attivissimo, A. Di Nisio, and M. Spadavecchia, "A low-cost tool for power quality monitoring in home automation systems," in *Proc. IEEE Instrum. Meas. Technol. Conf. (I2MTC)*, 2011, pp. 1–6.
- [9] Texas Instruments, "ADS131M02: 2-channel, simultaneously sampling, 24-bit, delta-sigma ADC," Datasheet SBAS865B, Sep. 2020. [Online]. Available: <https://www.ti.com/product/ADS131M02>
- [10] Zhongce Transformer Co., "ZMPT101B precision miniature voltage transformer," Component Datasheet, 2018.
- [11] A. Augustin, J. Yi, T. Clausen, and W. M. Townsley, "A study of LoRa: Long range and low power networks for the internet of things," *Sensors*, vol. 16, no. 9, p. 1466, Sep. 2016.
- [12] U. Raza, P. Kulkarni, and M. Sooriyabandara, "Low power wide area networks: An overview," *IEEE Commun. Surv. Tut.*, vol. 19, no. 2, pp. 855–873, 2nd Qtr. 2017.
- [13] M. Bor, J. Vidler, and U. Roedig, "LoRa for the internet of things," in *Proc. 13th Int. Workshop Mobile Ubiquitous Syst.: Comput., Netw. Services (EWSN)*, 2016.
- [14] Semtech Corporation, "SX1276/77/78/79: 137 MHz to 1020 MHz low power long range transceiver," Datasheet Rev. 7, 2020. [Online]. Available: <https://www.semtech.com/products/wireless-rf/lora-core/sx1276>
- [15] M. Dworkin, "Recommendation for block cipher modes of operation: Methods and techniques," NIST Special Publication 800-38A, Dec. 2001. [Online]. Available: <https://doi.org/10.6028/NIST.SP.800-38A>
- [16] LoRa Alliance, "LoRaWAN™ 1.0.4 specification," Oct. 2020. [Online]. Available: https://lora-alliance.org/resource_hub/lorawan-104-specification-package/
- [17] G. P. Hancke and M. G. Kuhn, "An RFID distance bounding protocol," in *Proc. 1st Int. Conf. Security Privacy Emerging Areas Commun. Netw. (SecureComm)*, 2005, pp. 67–73.
- [18] R. David *et al.*, "TensorFlow Lite Micro: Embedded machine learning for TinyML systems," *Proc. Mach. Learn. Syst. (MLSys)*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.08678>
- [19] P. Warden and D. Situnayake, *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*. Sebastopol, CA, USA: O'Reilly Media, 2019.
- [20] ARM Limited, "CMSIS-DSP software library," Documentation Version 1.15.0, 2024. [Online]. Available: https://arm-software.github.io/CMSIS_5/DSP/html/index.html
- [21] J. Gromes, "RadioLib: Universal wireless communication library for Arduino," GitHub, 2024. [Online]. Available: <https://github.com/jgromes/RadioLib>
- [22] IEEE, "IEEE standard definitions for the measurement of electric power quantities under sinusoidal, nonsinusoidal, balanced, or unbalanced conditions," IEEE Std 1459-2010, Feb. 2010.
- [23] STMicroelectronics, "STM32F411xC/xE reference manual," RM0383 Rev. 4, 2023. [Online]. Available: https://www.st.com/resource/en/reference_manual/rm0383-stm32f411xc-advanced-armbased-32bit-mcus-stmicroelectronics.pdf